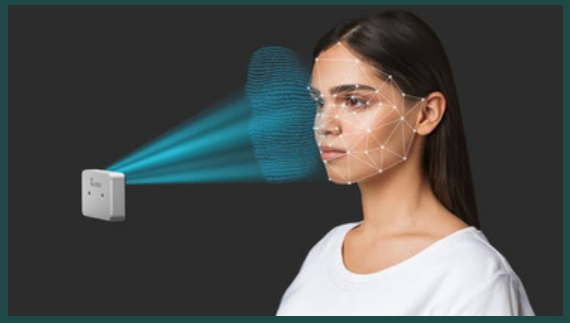
Achievements / Impact:

- Secured Smart Access
- Automated Irrigation System
- Modular & Scalable Design
- Real-time Monitoring
- Energy-efficient Automation

How it works:

1. Camera captures the face of a person at the door.
2. Cv2,numpy,pickles and os python libraries process the image and recognize the face using the LBPH algorithm.
4. Arduino unlocks the door using a servo motor and displays a welcome message.
5. Sensors monitor temperature, humidity, soil moisture, rain, light, and motion.
6. Based on sensor data, the system:
   - Waters plants
   - Turns on lights
   - Sends alerts
   - Auto-locks door
7. Can be controlled via app or voice assistant with smart modes: Home, Away, and Night.



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SMART HOME WITH FACE ID

2024-2025

About the Project:

A smart home system that uses facial recognition for secure access and automates various functions using environmental sensors and IoT.

Objective:

Enhance security and automation using AI, Arduino, and real-time sensors.

Key Features:

- Face ID Door Lock
- Rain Detection & Auto Window Close
- Smart Garden Irrigation
- Energy-Saving Lighting
- Intruder Detection



Components:

Main Hardware:

- Arduino
- DHT22 (Temp & Humidity)
- Soil Moisture Sensor
- HC-05 Bluetooth
- Servo Motor
- OLED & LCD Displays
- Light Sensor (LDR)
- Relay & Water Pump
- Ultrasonic Sensor
- Rain Sensor

Software:

- Python + OpenCV
- LBPH Algorithm for Face Recognition